



Physical Optics (Diffraction)

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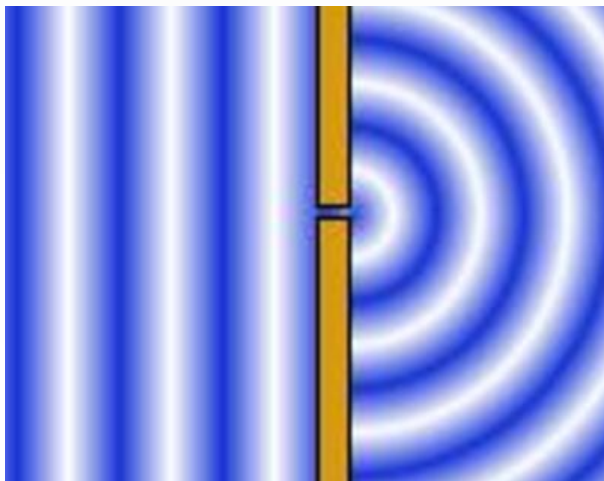
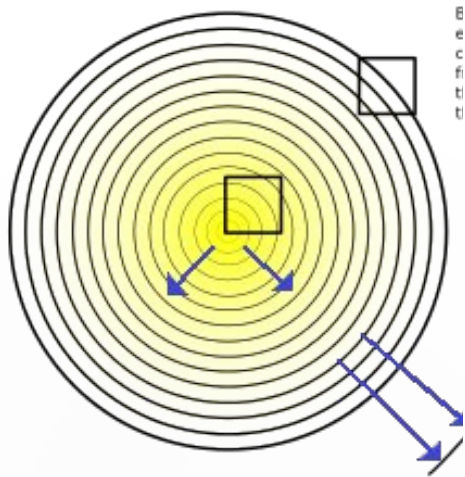
Diffraction is a term that defines the phenomena of waves interacting with particles. In this piece, we discuss diffraction of light waves, from diffraction examples to light diffraction equations.

What Is Diffraction?

We classically think of light as always traveling in straight lines, but when light waves pass near a barrier they tend to bend around that barrier and become spread out. The definition of diffraction is the spreading of waves as they pass through or around an obstacle. More specifically when applied to light, diffraction of light occurs when a light wave passes by a corner or through an opening or slit that is physically the approximate size of, or even smaller than that light's wavelength.

- A very simple demonstration of diffraction of waves can be conducted by holding your hand in front of a light source and slowly closing two fingers while observing the light transmitted between them. As the fingers approach each other and come very close together, you begin to see a series of dark lines parallel to the fingers. The parallel lines are actually diffraction patterns.
- A good example of this is the diffraction of sunlight by clouds that we often refer to as a silver lining, with a beautiful sunset over the ocean.
- We can often observe pastel shades of blue, pink, purple, and green in clouds that are generated when light is diffracted from water droplets in the clouds. The amount of diffraction depends on the wavelength of light, with longer wavelengths being diffracted at a greater angle than shorter ones

Schematically, the direction of movement of wave propagation can be depicted as lines or rays emanating from the source of the wave and perpendicular to the **wavefront** (see Figure 2). Close to the source, wavefronts will appear curved since they propagate in all directions from the source point. Far from the source, any small section of these wavefronts will appear much more flattened, and the rays are represented as virtually parallel to each other.



One might expect that when a traveling wavefront encounters a barrier with a small opening, the unobstructed portion of the wave will continue in the same straight-line direction the wave had before reaching the opening. This is not the case, however, due to diffraction. When a wavefront passes a small opening in a barrier, the wave on the other side of the barrier acts as though the opening itself is the source of the radiation: the waves once again spread out with semi-circular propagation (see Figures 3 & 4). In the case of light, the opening has the effect of **bending** the direction of the light's rays. Diffraction can happen for all types of waves, including electromagnetic radiation, sound, and water waves.



Two factors are important in diffraction

- the wavelength
- the size of the barrier's opening
- The position of the source and screen.

Diffraction happens dramatically *only* when the size of the aperture is small compared to the wavelength. Any aperture smaller than the wavelength can be considered a **point source** and will produce diffraction in the wave propagation past the barrier.

On the other hand, If the opening is large compared to the wavelength, most wave propagation is unchanged. Diffraction does occur, however, only near the edges of a barrier with a large opening.

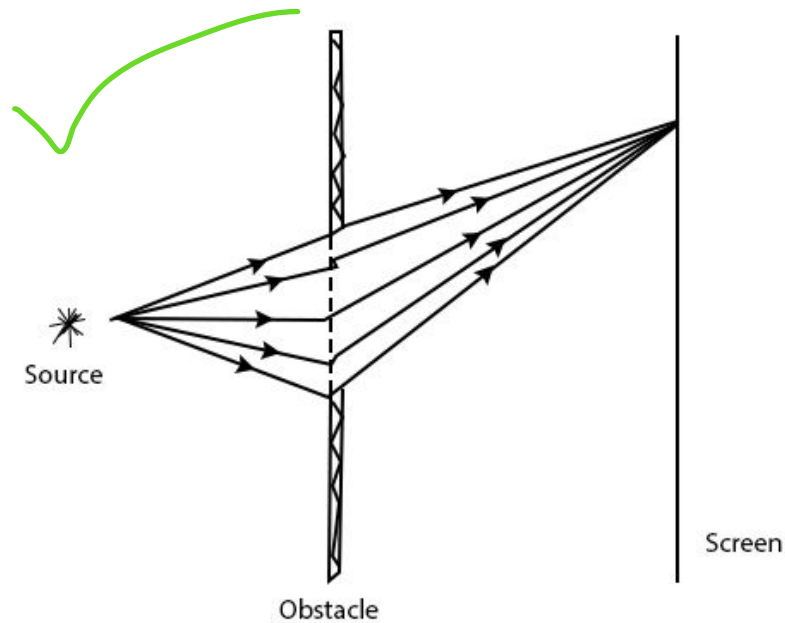
This explains **why we can hear a person speak around a corner but can't see them**. Sound waves are *very large* compared to the opening formed by the corner; the opening will serve as a point source for the sound. The wavelength of visual light is *vastly smaller* than this opening, so the diffraction of light is virtually non-existent in this case.



Historically, light radiation was first considered to be propagated only by streams of tiny particles called photons. If that were the case, however, one would expect that photons passing even a small aperture would continue in the same straight-line path they exhibited before the aperture. There was no reason to believe the trajectories of the particles would bend. The nature of light in terms of it being a particle versus a wave was one of the most contested debates in the history of science. Isaac Newton promoted the particle theory, which contrasted with that of Thomas Young, who believed light behaved as a wave. The observation that the diffraction of light is dependent on wavelength, as described above, supports the wave theory of electromagnetic radiation.

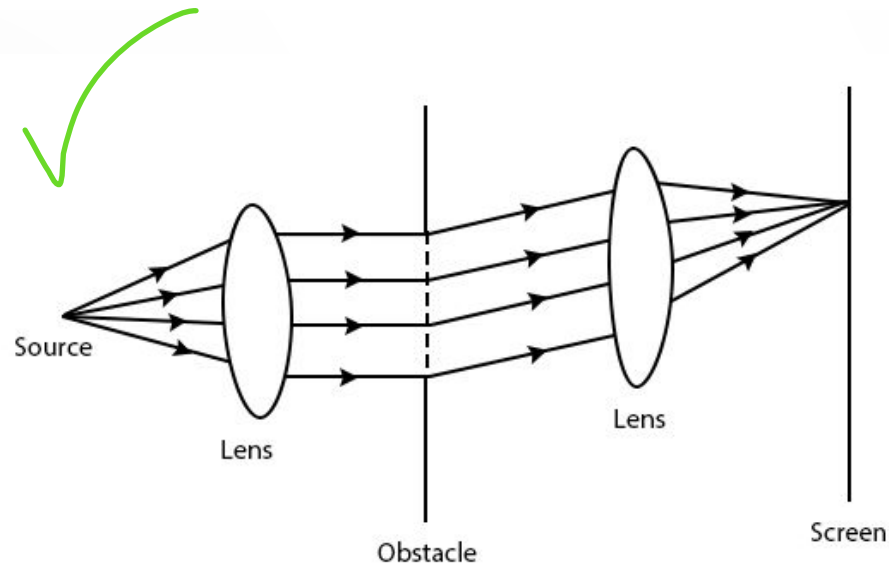
Fresnel Diffraction

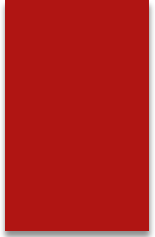
Fresnel diffraction refers to the phenomenon of light waves encountering an obstacle or passing through an aperture at finite distances. It occurs when the distances between the diffracting object, the light source, and the observation plane are comparable to the wavelength of light. Fresnel diffraction was initially observed and described by Augustin-Jean Fresnel, a French physicist, in the early 1800s.



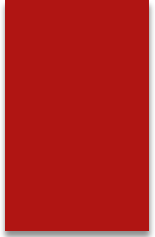
Fraunhofer Diffraction

Fraunhofer diffraction is a specific type of diffraction that occurs when light waves pass through an aperture or encounters an obstacle at infinite distances. In this diffraction, the observation point is located in the far-field region, relatively far away from the aperture or object, so that the incoming and outgoing waves are effectively planar. Since the source is at infinity, the rays of light which pass through the slit are parallel rays of light. In order to make these parallel rays to focus on the screen, a converging lens is used. It is named after the German physicist Joseph von Fraunhofer, who first observed the phenomenon in the early 19th century.





Fraunhofer diffraction	Fresnel diffraction
Source and screen are at <u>infinite</u> distances from slit.	Source and screen are at <u>finite</u> distances from slit.
Incident wavefront on the aperture is <u>plane</u> .	Incident wavefront on the aperture is either <u>spherical or cylindrical</u> .
The diffracted wavefront is <u>plane</u>	The diffracted wavefront is either <u>spherical or cylindrical</u>
<u>Two biconvex lenses</u> are required to study diffraction in laboratory	<u>No lenses</u> are required to study diffraction in the laboratory
Mathematical treatment is <u>easy</u>	Mathematical treatment is <u>complicated</u>
It has <u>many applications</u> in designing the optical instruments	It has <u>less applications</u> in designing the optical instruments
The maximas and minimas are <u>well defined</u>	The maximas and minimas are <u>not well defined</u> .



Three common types of diffraction are:

- **Single-slit diffraction.** The obstacle in the wave's path has a single opening that may be smaller or somewhat larger than the wavelength.
- **Double-slit diffraction.** The obstacle has two very small openings spaced a certain distance apart.
- **Diffraction grating.** The obstacle has multiple small openings or grooves evenly spaced apart. The number of slits per unit length is the grating density.

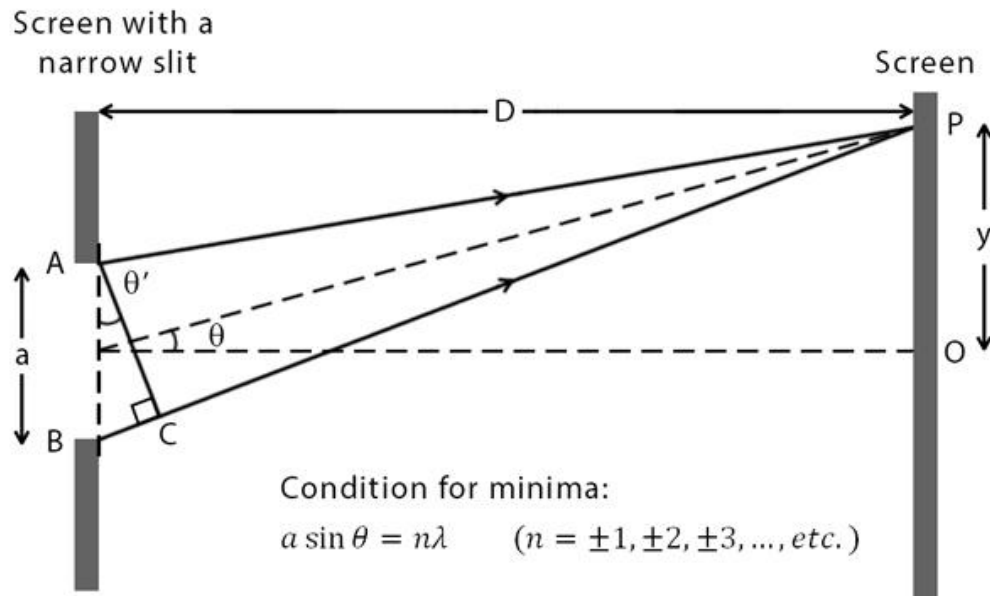
For light waves that are in phase and monochromatic, meaning they are coherent and have the same wavelength, these experimental set-ups produce diffraction patterns that can be visualized on a perpendicular detector screen at a distance from the openings.

In each case, the point sources at the openings will produce multiple waves that undergo interference. **Constructive interference** happens when wave amplitudes sum together to produce intensity maxima (bright lines on the screen).

Destructive interference happens when wave amplitudes cancel each other to form intensity minima (dark lines). Figure 8 shows where the positions of intensity maxima and minima will occur on a detector screen for single-slit diffraction.

Single Slit Diffraction Equation

In order to study the diffraction pattern on a screen, the single-slit experiment is employed. Consider a monochromatic source of light that passes through a slit AB of width a as shown in the figure. At point P on the screen, the secondary waves interfere destructively and produce a dark fringe. Let D be the distance between the slit and the screen, and y be the distance between point P and point O, the center of the screen. AC is perpendicular to BP. Let θ be the angle of diffraction, and θ' is the angle BAC.



We assume that the screen is at a considerable distance from the slit, i.e., $D \gg a$.
Hence,

$$\theta = \theta'$$

And,

$$\sin \theta \approx \tan \theta \approx \theta = y/D$$

The path difference between the two rays AP and BP is given by,

$$\Delta = BP - AP = BC$$

In the right-angled triangle BCA,

$$\sin \theta' = \sin \theta = BC/BA$$

$$BC = BA \sin \theta = a \sin \theta$$

Therefore,

$$\Delta = a \sin \theta$$

Diffraction Minima

The condition for minima or dark fringe is,

$$\text{or, } \Delta = n\lambda \text{ (} n=\pm 1, \pm 2, \pm 3, \dots, \text{etc.)}$$

For small θ ,

$$\mathbf{\sin \theta \approx \theta}$$

$$\Rightarrow \lambda = a \sin \theta \approx a\theta$$

$$\Rightarrow \theta = y/D = \lambda/a$$

$$\Rightarrow \mathbf{y} = \lambda D/a$$

$$\text{or, } a \sin \theta = n\lambda$$

$$\text{or, } ay/D = n\lambda$$

$$\text{or, } y_n = n\lambda D/a$$

This equation gives the distance of the n-th dark fringe from the center.

Diffraction Maxima

The condition for maxima or bright fringe is,

$$\text{or, } \Delta = (n+1/2)\lambda \text{ (} n=\pm 1, \pm 2, \pm 3, \dots, \text{etc.)}$$

$$\text{or, } a \sin \theta = (n+1/2)\lambda$$

$$\text{or, } ay/D = (n+1/2)\lambda$$

$$\text{or, } y_n = (n+1/2)\lambda D/a$$

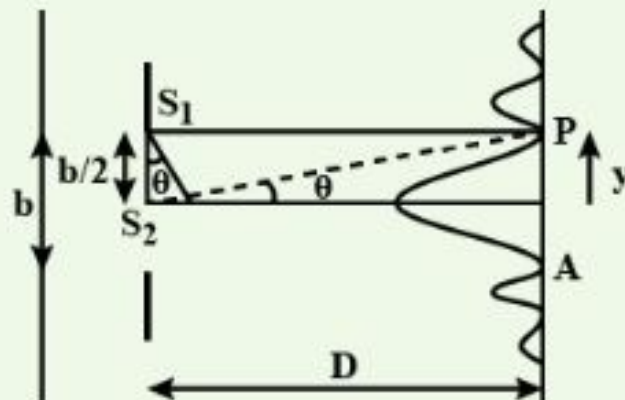
The maxima lie between the minima and the width of the central maximum is simply the distance between the 1st order minima from the center of the screen on both sides of the center

1st order minima from the centre of the screen

$$\Rightarrow y = \lambda D/a$$

$$\Rightarrow \text{Width of central maximum} = 2\lambda D/a$$

SOLUTION



Let the width of the slit be through which diffraction pattern is obtained be b and λ be the wavelength of light used.

The first minima of diffraction pattern is formed at point P on one side and at A of other side.

Path difference between the light rays reaching at point P $\Delta x = S_2P - S_1P$

$$\therefore \Delta x = \frac{b}{2} \sin \theta$$

For first minima, path difference must be equal to $\frac{\lambda}{2}$

$$\therefore \frac{b}{2} \sin \theta = \frac{\lambda}{2}$$

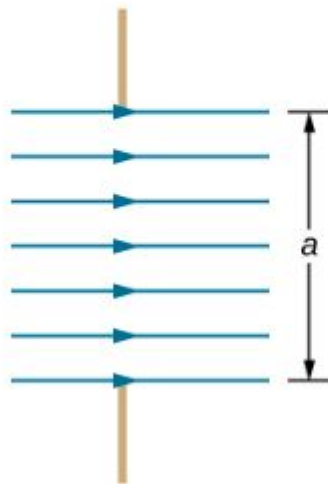
$$\text{OR } b \sin \theta = \lambda$$

As θ is very small, thus $\sin \theta \approx \tan \theta = \frac{y}{D}$

$$\text{OR } b \times \frac{y}{D} = \lambda \implies y = \frac{\lambda D}{b}$$

$$\text{Width of central maxima } PA = 2y = \frac{2\lambda D}{b}$$

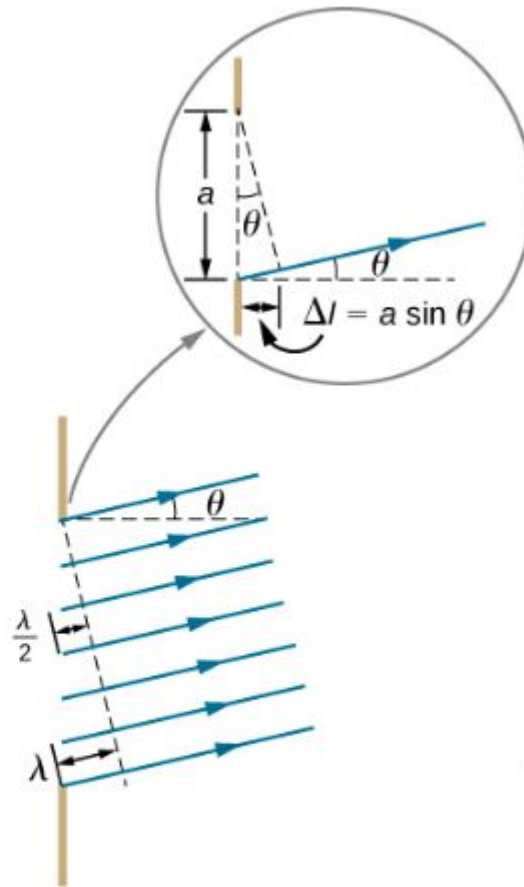
For understanding



$$\theta = 0$$

Bright

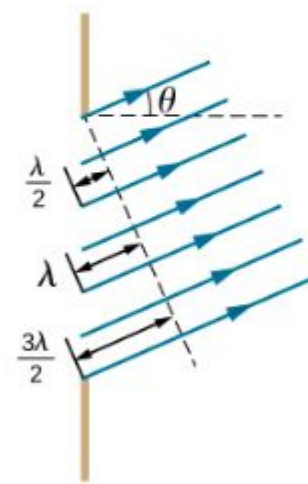
(a)



$$\sin \theta = \frac{\lambda}{a}$$

Dark

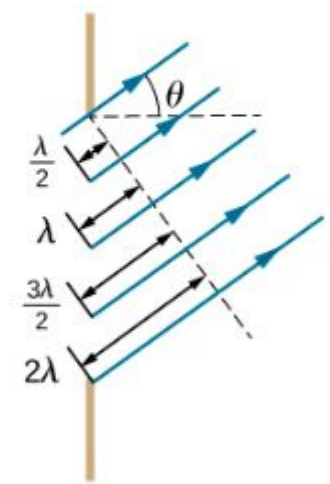
(b)



$$\sin \theta = \frac{3\lambda}{2a}$$

Bright

(c)



$$\sin \theta = \frac{2\lambda}{a}$$

Dark

(d)

According to [Huygens's principle](#), every part of the wave front in the slit emits wavelets, as we discussed in [The Nature of Light](#). These are like rays that start out in phase and head in all directions. (Each ray is perpendicular to the wave front of a wavelet.) Assuming the screen is very far away compared with the size of the slit, rays heading toward a common destination are nearly parallel. When they travel straight ahead, as in part (a) of the figure, they remain in phase, and we observe a central maximum. However, when rays travel at an angle θ relative to the original direction of the beam, each ray travels a different distance to a common location, and they can arrive in or out of phase. In part (b), the ray from the bottom travels a distance of one wavelength λ farther than the ray from the top. Thus, a ray from the center travels a distance $\lambda/2$ less than the one at the bottom edge of the slit, arrives out of phase, and interferes destructively. A ray from slightly above the center and one from slightly above the bottom also cancel one another. In fact, each ray from the slit interferes destructively with another ray. In other words, a pair-wise cancellation of all rays results in a dark minimum in intensity at this angle. By symmetry, another minimum occurs at the same angle to the right of the incident direction (toward the bottom of the figure) of the light.

At the larger angle shown in part (c), the path lengths differ by $3\lambda/2$ for rays from the top and bottom of the slit. One Ray travels a distance λ different from the Ray from the bottom and arrives in phase, interfering constructively. Two rays, each from slightly above those two, also add constructively. Most rays from the slit have another ray to interfere with constructively, and a maximum in intensity occurs at this angle. However, not all rays interfere constructively for this situation, so the maximum is not as intense as the central maximum. Finally, in part (d), the angle shown is large enough to produce a second minimum. As seen in the figure, the difference in path length for rays from either side of the slit is $a\sin\theta$, and we see that a destructive minimum is obtained when this distance is an integral multiple of the wavelength.

The most modern-day applications of slit interference use not just two slits but many, approaching infinity for practical purposes. The key optical element is called a diffraction grating, an important tool in optical analysis. A prime example is an optical element called a diffraction grating. A diffraction grating can be manufactured by carving glass with a sharp tool in a large number of precisely positioned parallel lines, with untouched regions acting like slits

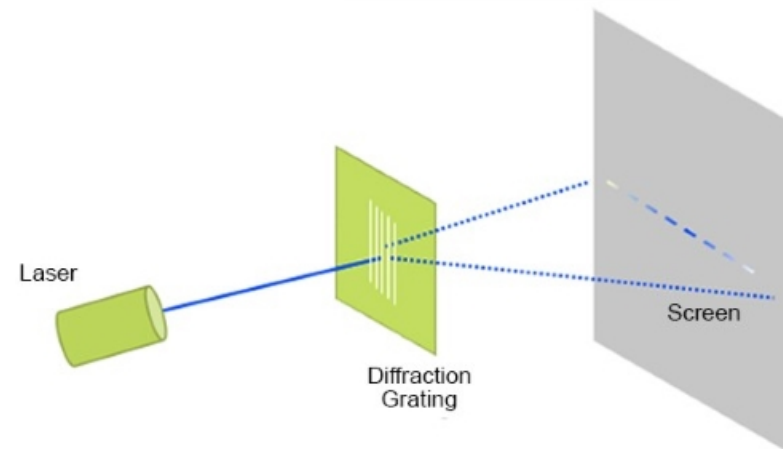
A diffraction grating defines an optical component with a periodic structure that splits the light into various beams that travel in different directions. It is an alternative way to observe spectra other than a prism. Generally, when light is incident on the grating, the split light will have maxima at an angle θ . The formula for diffraction grating is used to calculate the angle.

Diffraction grating formula

Where,

$$\lambda_n = d \sin \theta_n$$

- n is the order of grating,
- d is the distance between two fringes or spectra
- λ is the wavelength of light
- θ is the angle to maxima



Question 1: A diffraction grating is of width 5 cm and produces a deviation of 30° in the second-order with the light of wavelength 580 nm. Find the slit spacing.

Solution:

Given: Angle $\theta = 30^\circ$, order $n = 2$, wavelength $\lambda = 580 \text{ nm}$.

The slit spacing is given by,

$$\lambda n = d \sin \theta_n$$

$$d = 2 \times 580 \times 10^{-9} / \sin 30^\circ$$

$$d = 2.320 \times 10^{-6} \text{ m}$$



A diffraction grating is an optical component that is used to disperse light into its constituent colors or wavelengths. It consists of a large number of equally spaced, parallel slits or rulings that are closely packed together. When light passes through a diffraction grating, it undergoes a phenomenon called diffraction, which results in the separation of light into its various spectral components.

Introduction to Diffraction Grating

A diffraction grating is a powerful tool in optics and spectroscopy. It is designed to separate incoming light into its individual wavelengths or colors, which is essential for applications such as spectroscopy, where the analysis of light's spectral composition is crucial. Diffraction gratings are commonly used in devices like spectrometers, monochromators, and other optical instruments for this purpose.

How Diffraction Gratings Formula Work?

Diffraction gratings work based on the principle of diffraction, which is the bending of light as it encounters an obstacle or aperture. When light passes through the narrow slits or rulings of a diffraction grating, it undergoes diffraction, causing the light waves to interfere with each other constructively and destructively. This interference pattern results in the dispersion of light into its various wavelengths.



The key factors that determine the characteristics of the diffraction pattern produced by a grating are:

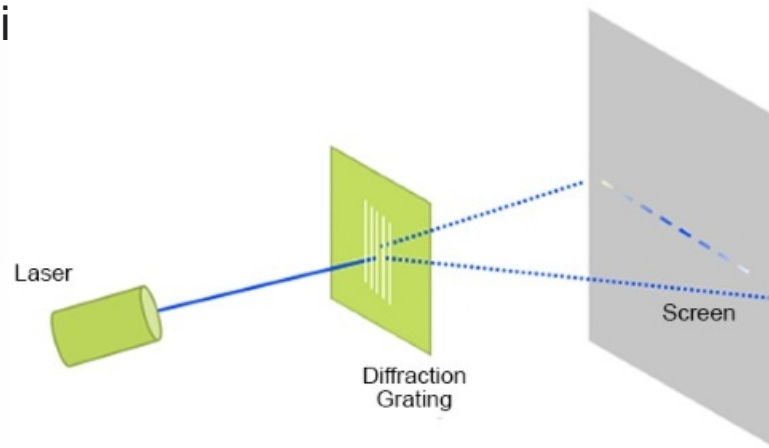
- Wavelength of light (λ): The wavelength of the incident light is a crucial factor. Different wavelengths will be separated by different angles, allowing the grating to act as a spectral disperser.
- Spacing of the grating (d): The distance between adjacent rulings or slits on the grating is known as the grating spacing. It plays a significant role in determining the angles at which different wavelengths are diffracted.
- Order of diffraction (m): Diffraction can occur at different orders, labeled as m . The zeroth order ($m = 0$) corresponds to un-diffracted light, while positive and negative orders represent different orders of diffraction. Higher orders result in greater angular separation between wavelengths.

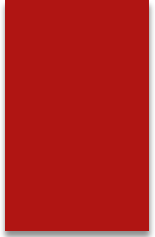
3.The Diffraction Grating Equation:

The diffraction grating equation relates the angle of diffraction (θ), the wavelength of light (λ), the grating spacing (d), and the order of diffraction (m). The equation is given as:

$$m\lambda = d \sin(\theta)$$

- m : The order of diffraction. It can be a positive or negative integer, including zero. The zeroth order corresponds to the central undiffracted beam.
- λ : The wavelength of the incident light, usually measured in nanometers (nm) or angstroms (Å).
- d : The grating spacing, which is the distance between adjacent rulings on the grating. It is typically measured in units of length, such as meters or millimeters.
- θ : The angle of diffraction, which is the angle between the incident light and the direction of the diffracted li





Applications of Diffraction Grating

Diffraction gratings have a wide range of applications across various fields. Some of the notable applications include:

- **Spectroscopy:** Diffraction gratings are fundamental components in spectrometers used for chemical analysis, astronomy, and materials characterization. They enable the separation and measurement of the spectral lines emitted or absorbed by different substances.
- **Wavelength Selection:** In lasers, diffraction gratings formula are used to select specific wavelengths, allowing for the tuning of laser output. This is crucial in applications like telecommunications and medical devices.
- **Color Separation:** In optics and photography, diffraction gratings are used to separate colors, as seen in some camera lenses and optical filters.
- **Astronomy:** Gratings are used in astronomical spectrographs to analyze the light emitted or absorbed by celestial objects, helping scientists determine their chemical composition, temperature, and other properties.
- **Light Analysis:** In physics experiments, gratings are used to study the behavior of light and validate theories related to wave optics.